

TARA: Threat analysis and risk assessment methods

It's a method for analyzing threats and risks present in automotive digital security components
The methodology is divided in 7 phases

1) Asset identification

An asset is an object that has value, or contributes to the whole value

It has one or more cybersecurity properties, if compromised, can lead to damage scenarios

These properties are: Confidentiality, Availability, Integrity

Phase output: • damage scenario identification (consequences of cysec properties violation and who is involved)

Input: • identification of assets that can lead to a damage scenario if compromised

2) Threat scenario identification

Output: • threat scenario identification, target assets, compromised properties, causes of compromise

• One Damage scenario can correspond to various threat scenario and vice-versa

3) Impact rating

Output: evaluation of Damage Scenario

Impact categories are related to road users (Safety, Financial, Operational, Privacy)

Impact ratings: Severe, Major, Moderate, Negligible

4) Attack path analysis

Output: determines attack paths which leads to a threat scenarios

• Deduce attack paths from trees, ...

• Build path from known vulnerabilities

5) Attack feasibility rating

Output: for each attack path, its feasibility (how much is hard to complete the attack)

Can be High, Medium, Low, Very Low. May be based on CVSS or attack potential

6) Risk value determination

Output: for each threat scenario, risk obtained by combining impact of damage scenarios and feasibility of attack paths related to that threat scenario

An example: Risk matrix: impact rating * Feasibility rate

7) Risk treatment decision

Output: for each threat scenario, a treatment option

• Avoid risk: remove risk source, drop risky activity

• Reduce risk

• Sharing risk: e.g. insurance

• Retaining risk